Code No. 8401

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (AI&ML) IV - Semester (MAIN & BACKLOG) Examinations AUGUST-2024

ARTIFICAL INTELLIGENCE

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** is compulsory and answer any **four questions** from the remaining questions (2- 7), each carry **12** Marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any, may suitably be assumed.***

		Marks	BT	CO
1. a)	What is the Turing Test.	2	2	1
b)	Write the PEAS description for the medical diagnosis system.	2	2	1
c)	When can you say an AI agent is rational.	2	2	2
d)	Write the role of a Shell of an Expert System.	2	1	3
e)	Distinguish between crisp set and fuzzy set.	2	2	2
f)	List out the applications where Neural networks are used.	2	1	2
2. a)	Define and illustrate the idea of agent, agent function and agent program by using vacuum cleaner problem.	6	2	1
b)	Apply the depth limited DFS for the following 8-puzzle problem upto level 5	6		

Initial State

7	2	4
5		6
8	3	1

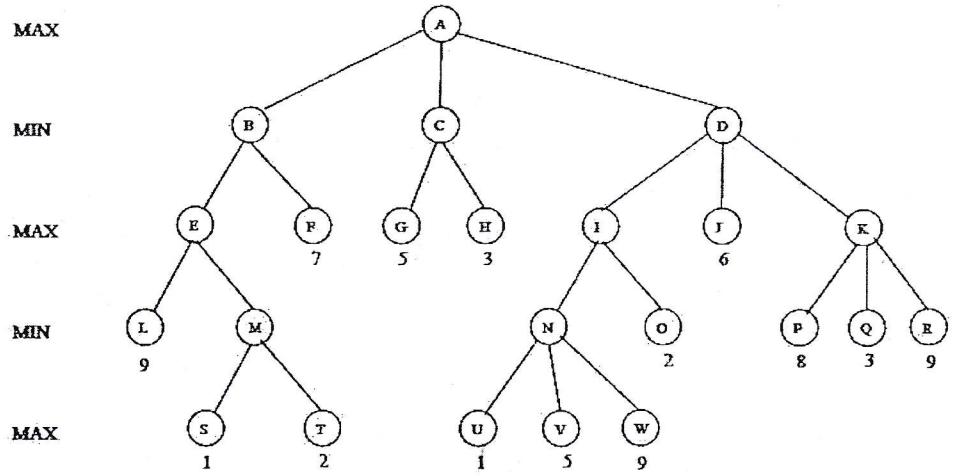
Goal State

	1	2
3	4	5
6	7	8

3. a)	Write A* algorithm and explain with an example.	6	2	2

(P.T.O.)

b) Explain how the concept of pruning will help in solving the following MINIMAX game efficiently. Also display the pruned tree. 6 3 2



4. a) Explain rules of inference in propositional calculus. 6 2 1

b) Represent the given sentences into FOPL and convert them into clausal form 6 3 2

(a) Marcus was a man
(b) Marcus was a Roman
(c) All men are people
(d) Caesar was a ruler.
(e) All Romans were either loyal to Caesar or hated him (or both)
(f) Everyone is loyal to someone
(g) People only try to assassinate rulers they are not loyal to
(h) Marcus tried to assassinate Caesar

5. a) What is an expert system. Explain the main components of an expert system. 6 2 1

b) Explain the variable elimination Algorithm used for inference in Bayesian Networks with example. 6 3 3

6. a) Discuss briefly about the applications of Neural Networks. Describe the architecture and learning rule of Perceptron. 6 2 2

b) Write a short notes on Hopfield Networks. 6 2 2

7. a) Assume a domain with three attributes A, B, and C. Each attribute has two possible values T and F. Given below is a set of instances. Calculate the information gain for the attributes A, B and C. Which attribute would be selected by the standard ID3 algorithm — 7 3 2

A	B	C	Target
T	T	T	Yes
T	T	F	NO
T	F	T	Yes
F	T	T	Yes
F	T	F	No
F	F	F	Yes

b) Compare forward versus backward reasoning. — 5 2 2

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